

The Fertility, Marriage, and Gender Equality Quandary

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Motivation

- Three trends underpin the grand gender convergence in the last century:
 1. [Falling fertility](#) (Guinnane 2011)
 2. [Declining marriage](#) (Stevenson and Wolfers 2007)
 3. [Converging gender \(income\) gaps](#) (Goldin 2014)
- Many policies targeting individual outcomes
- Existing research often treats them in isolation or pairwise, lacking a unified framework

This Paper

1. Document a **three-way trade-off**: simultaneously achieving high fertility, low single parenthood, and gender income equality is statistically unlikely
2. Develop a **unified model** of marriage, fertility, and female labor supply to explain the empirical facts
3. Calibrate to Mexico (1990–2015), conduct decomposition and prediction

Key Findings

1. Mutual information analysis reveals **synergy**: the three outcomes should be analyzed jointly
2. Traditional policies cannot resolve the trade-off
3. **Redistributing childcare** to husbands can under two conditions
4. Quantitative results on Mexico from 1990 to 2015:
 - Gender-neutral TFP explains half of declining fertility and marriage
 - Gender-biased TFP and education gap reversal drive income convergence
5. Dynamic propagation through gendered impacts of single parenthood

Literature

- Pairwise studies of fertility, marriage, and gender gaps (Galor & Weil 1996, Regalia & Ríos-Rull 2001, Greenwood et al. 2016)
→ Establish three-way trade-off; unified framework
- Gender-biased technical change (Greenwood et al. 2005, Ngai & Petrongolo 2017)
→ Show gender-neutral TFP also drives convergence
- Boys' disadvantage in single-parent families (Autor et al. 2019)
→ Embed in macro model with dynamic propagation
- Policy evaluation in family economics (Doepke & Kindermann 2019)
→ Identify policy frontier; childcare redistribution as remedy

Roadmap

- Empirical findings
- Model
- Policy analysis
- Quantitative analysis
- Dynamic extension
- Conclusion

Motivating Facts

Empirical Approach

1. Create panel datasets with fertility, dual parenthood, and gender income gap
2. Create dummy variables at sample medians:
 - High fertility (F)
 - Dual parenthood (M)
 - Gender income equality (G)
3. Visualize intersections using Venn diagrams
4. Compute mutual information to characterize dependency structure

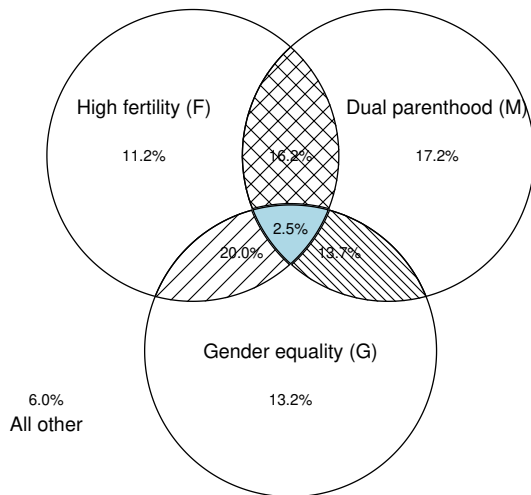
Dartboard approach: random benchmark is $0.5 \times 0.5 \times 0.5 = 12.5\%$

OECD Sample

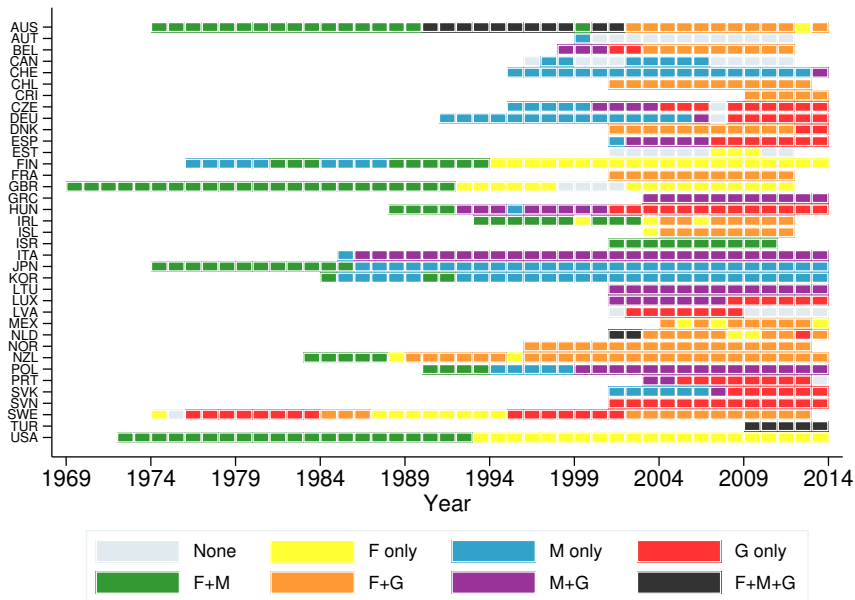
37 OECD countries, 1970–2014
721 observations

- Total fertility rate
- Out-of-wedlock birth share
- Gender median earnings gap

Trinity share: 2.5%



OECD Sample – Transition Paths



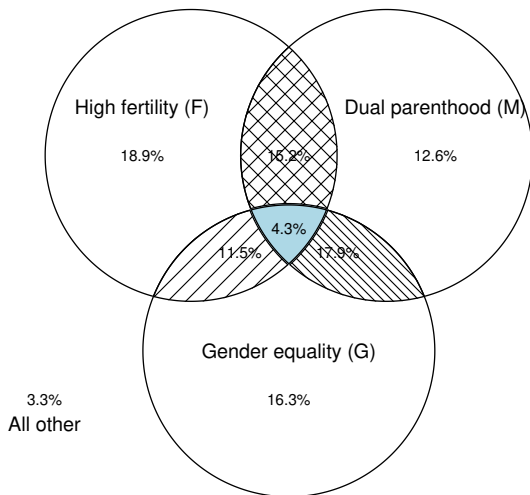
World Sample

95 countries, 1990–2015

1,130 observations

- Total fertility rate
- Children with both parents
- Female labor income share

Trinity share: 4.3%

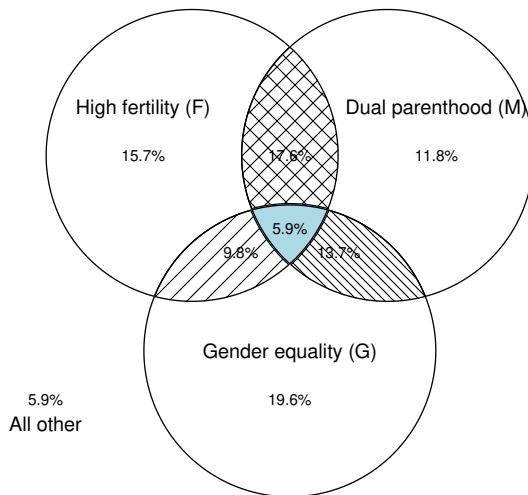


U.S. Sample (States)

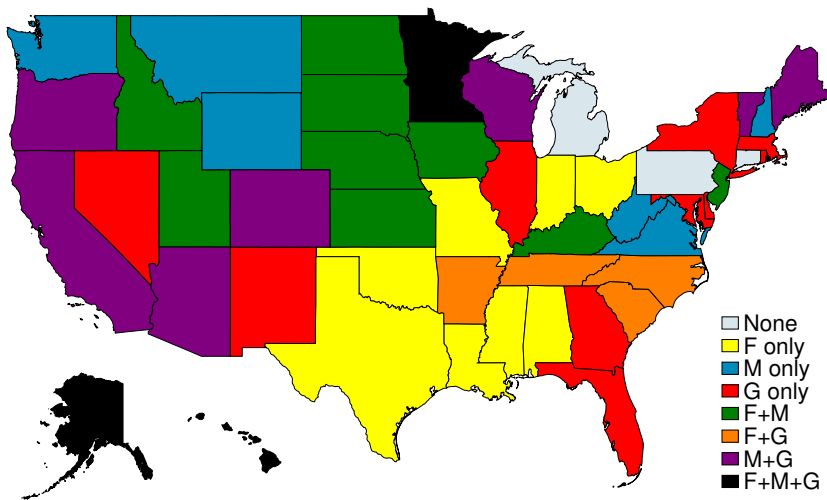
50 states + D.C., 2023

- Births per 1000 women
- Single-mother household share
- Gender earnings gap

Trinity share: 5.9%



U.S. Sample (States) – Geographic Distribution



Mutual Information Analysis

- Key finding: **Synergy** — dependence between any two variables *increases* when conditioning on the third
- Implication: analyzing pairs misses dependency structure

Sample	Total Corr.	Interaction	Interpretation
OECD	0.165	−0.035***	Synergy
World	0.169	−0.026***	Synergy
U.S. States	0.112	−0.007	Synergy

Questions

1. What explains the three-way trade-off?
2. Are there policies that can help escape the trade-off?
3. What role does technological progress play?

Model

Basic Setup

- Building on Becker (1973), Choo and Siow (2006)
- Individuals with gender $g \in \{\sigma, \varphi\}$ and preference

$$u(c, n) = \left((1 - \beta) \cdot c^{\frac{\rho-1}{\rho}} + \beta \cdot n^{\frac{\rho-1}{\rho}} \right)^{\frac{\rho}{\rho-1}} \quad (1)$$

where $\rho > 1$ so that $u(c, 0)$ is well-defined

- Homogeneous wage within gender: w^{σ} and w^{φ}
- Gender wage gap: $\Gamma^w = w^{\sigma} / w^{\varphi}$

Marriage and Fertility – Men

- If single, men consume labor income but have no children

$$V^{\sigma^{\uparrow},s} = u(w^{\sigma^{\uparrow}}, 0) \quad (2)$$

- Once married, husbands transfer α share of income to wives

$$V^{\sigma^{\uparrow},m} = u((1 - \alpha)w^{\sigma^{\uparrow}}, n^m) \cdot \lambda \quad (3)$$

- α is endogenous in marriage market equilibrium
- λ is the psychic benefit of marriage

Marriage and Fertility – Women

- Single women solve

$$V^{\text{♀},s} = \max_{c,l,n} u(c, n) \quad \text{s.t.} \quad c = w^{\text{♀}}l, \quad l = 1 - \chi n \quad (4)$$

- Married women solve

$$V^{\text{♀},m} = \max_{c,l,n} u(c, n) \cdot \lambda \quad \text{s.t.} \quad c = \alpha w^{\text{♂}} + w^{\text{♀}}l, \quad l = 1 - \chi n \quad (5)$$

- Fertility subject to wife's veto (Doepke & Kindermann 2019) $\implies$ wives determine marital fertility

Marriage Market Equilibrium

- Idiosyncratic taste shock ϵ for marriage, Fréchet with scale θ
- Marriage rates:

$$\mathcal{M}^{\sigma} = \frac{1}{1 + (V^{\sigma,s}/V^{\sigma,m})^{\theta}}, \quad \mathcal{M}^{\varphi} = \frac{1}{1 + (V^{\varphi,s}/V^{\varphi,m})^{\theta}} \quad (6)$$

- Equilibrium: $\mathcal{M}^{\sigma} = \mathcal{M}^{\varphi} = \mathcal{M}$
- (α, n^m) act as market-clearing “prices”

Aggregate Quantities

- Aggregate fertility and dual parenthood share:

$$n = \mathcal{M} \cdot n^m + (1 - \mathcal{M}) \cdot n^s, \quad \mathcal{D} = \frac{\mathcal{M} \cdot n^m}{n} \quad (7)$$

- Female labor supply:

$$l^{\ominus} = \mathcal{M} \cdot l^m + (1 - \mathcal{M}) \cdot l^s = 1 - \chi n \quad (8)$$

- Gender income gap:

$$\Gamma^y = \frac{y^{\sigma^{\nearrow}}}{y^{\ominus}} = \frac{\Gamma^w}{l^{\ominus}} \quad (9)$$

Equilibrium Characterization

- Lemma 1: For any wage pair $\{w^{\sigma}, w^{\varphi}\}$, there exists a unique equilibrium (α, n^m)
- Lemma 2: Gains from marriage: $V^{\varphi, m}/V^{\varphi, s} = \lambda \cdot (1 + \alpha\Gamma^w)$
- Equilibrium conditions reveal fundamental tension:

$$\mathcal{M} = \frac{(\lambda \cdot (1 + \alpha\Gamma^w))^{\theta}}{1 + (\lambda \cdot (1 + \alpha\Gamma^w))^{\theta}} \quad (10)$$

$$\Gamma^y = \Gamma^w / l^{\varphi} \quad (11)$$

$$l^{\varphi} = 1 - \chi n \quad (12)$$

Three-Way Trade-Off: Intuition

- High n + High $\mathcal{M}$: Low $l^{\text{♀}}$ requires large Γ^w to sustain marriage $\Rightarrow$ high Γ^y
- High n + Low Γ^y : Low $l^{\text{♀}}$ requires tiny Γ^w for equality $\Rightarrow$ low $\mathcal{M}$
- High $\mathcal{M}$ + Low Γ^y : Large Γ^w needed for marriage; offsetting requires high $l^{\text{♀}} \Rightarrow$ low n

Core mechanism: Women's disproportionate childcare burden ties fertility to gender income gaps through labor supply

Policy Analysis

Traditional Policy Effects

Policy	Fertility (n)	Marriage ($\mathcal{M}$)	Gender Gap (Γ^y)
Pro-marriage ($\uparrow \lambda$)	+	+	+ (widens)
Gender equality ($\uparrow w^{\text{♀}}$)	−	−	− (narrows)
Reduce child costs ($\downarrow \chi$)	+	+	+ (widens)

Key insight: None of these policies can simultaneously improve all three outcomes

Redistributing Childcare to Husbands

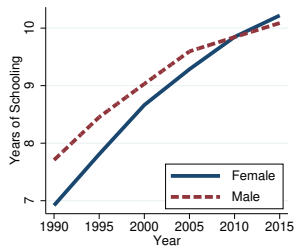
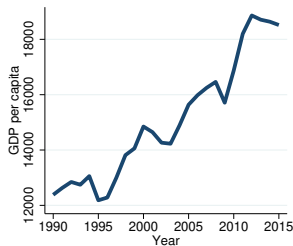
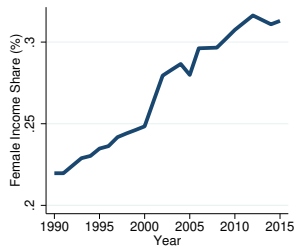
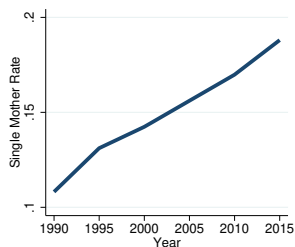
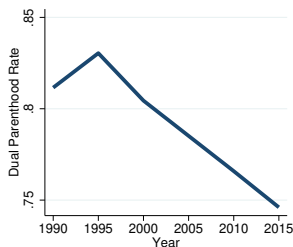
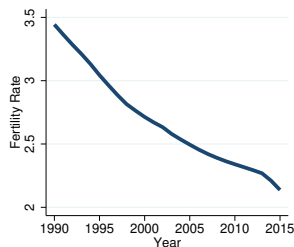
- Extend model: $\chi^{\text{♀}}$ for wives, $\chi^{\text{♂}}$ for husbands, $\chi = \chi^{\text{♀}} + \chi^{\text{♂}}$ for single mothers
- Wife's effective childcare cost: $\tilde{\chi}^{\text{♀}} = \frac{\alpha w^{\text{♂}} \chi^{\text{♂}} + w^{\text{♀}} \chi^{\text{♀}}}{\alpha w^{\text{♂}} + w^{\text{♀}}}$
- **Proposition:** Redistribution improves all three outcomes if:
 1. Wife's earnings exceed spousal transfers: $\alpha < w^{\text{♀}}/w^{\text{♂}}$
 2. Husband's utility gain from fertility exceeds consumption loss
- When both hold, redistribution is **Pareto-improving**

Policy Implications

- The trade-off may be partly an artifact of gendered childcare division
- Policies promoting paternal involvement (e.g., “use-it-or-lose-it” paternity leave) may help escape the trade-off
- **Caveat:** Success depends on the economic environment and social norms

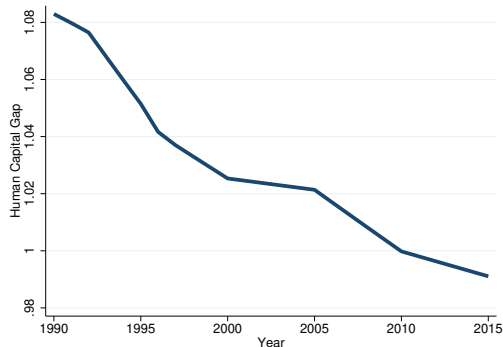
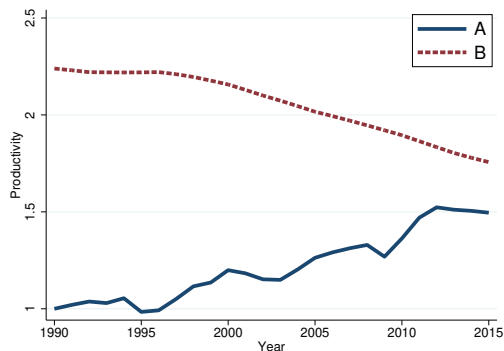
Quantitative Analysis

Mexico's Transition (1990–2015)



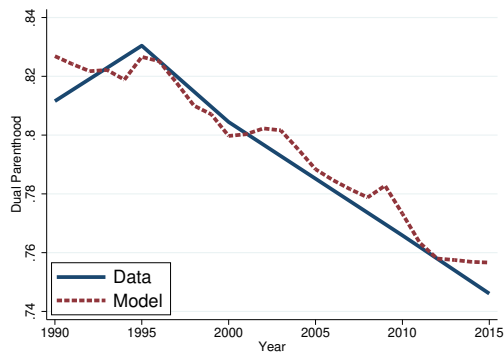
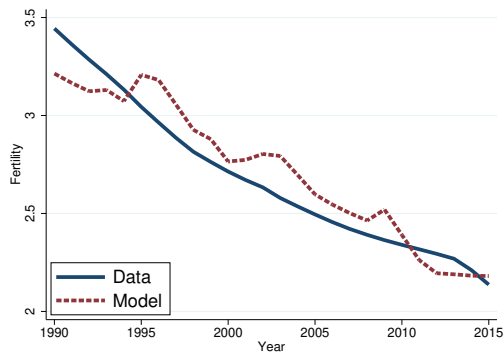
Calibration

- Decompose wages into three components:
 - $A_t = w_t^{\sigma}$: gender-neutral productivity
 - Γ_t^h : gender gap in human capital
 - $B_t = (w_t^{\sigma} / w_t^{\varphi}) / \Gamma_t^h$: gender-biased productivity



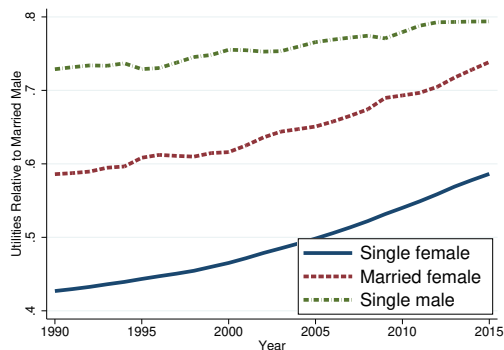
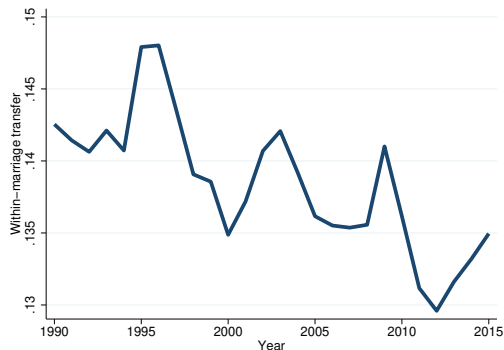
Model Fit

- Fix $\chi = 0.075$ (de La Croix and Doepke 2003) assuming $\chi^{\sigma} \approx 0$
- Calibrate: $\beta = 0.104$, $\rho = 1.5$, $\lambda = 1.03$, $\theta = 3.4$

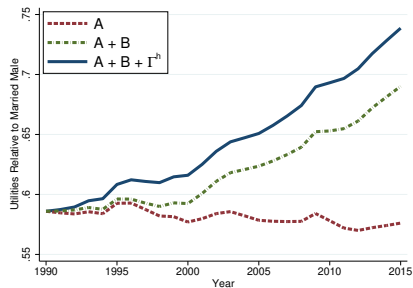
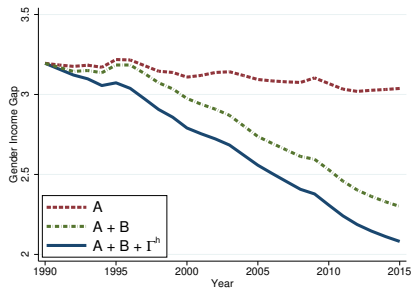
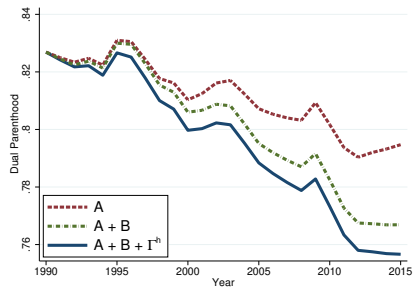
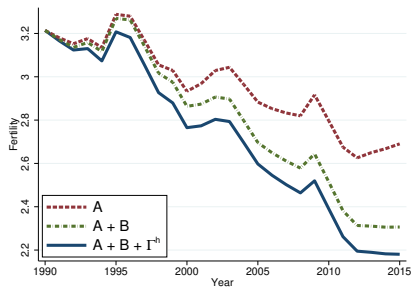


Equilibrium Outcomes

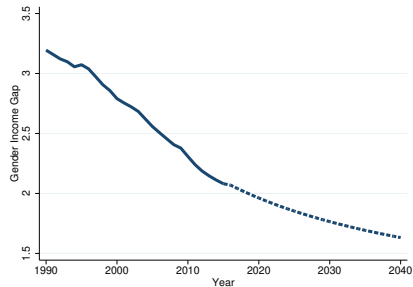
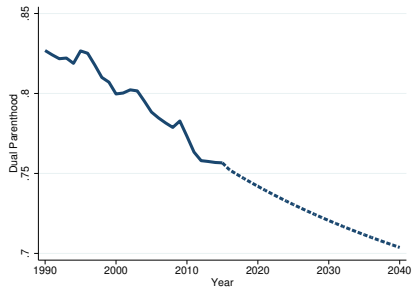
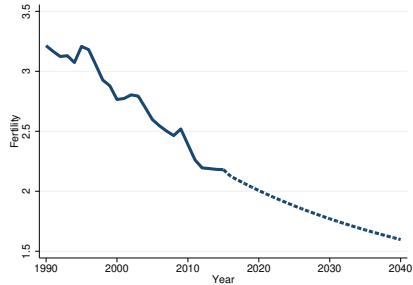
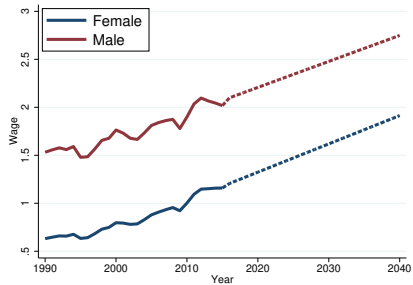
- Small reduction in α despite large drop in n^m as $w^{\text{♀}}/w^{\text{♂}}$ rises
- $\alpha \approx 0.14$ satisfies condition for wives to benefit; husband condition also satisfied $\implies$ redistribution would be Pareto-improving



Decomposition



Model-Based Prediction



Dynamic Extension

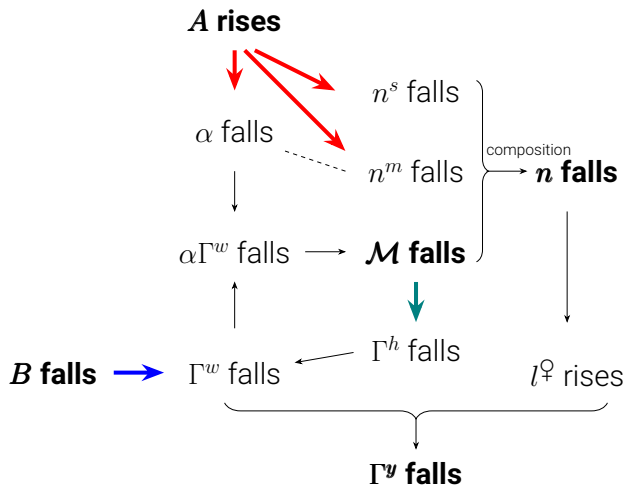
Endogenous Gender Human Capital Gap

- Single parenthood affects boys and girls differently
- Evidence: Bertrand & Pan 2013, Autor et al. 2019, Wasserman 2020, Reeves 2022
- Law of motion:

$$\Gamma_{t+1}^h = \mathcal{H}(\mathcal{M}_t) \quad \text{where } \mathcal{H}'(\mathcal{M}_t) > 0 \quad (13)$$

- Creates self-reinforcing feedback: declining marriage $\rightarrow$ lower male human capital $\rightarrow$ further marriage decline

Dynamic Mechanism



Red: A_t rises Blue: B_t falls Teal: Dynamic feedback

Conclusion

- A three-way trade-off with synergistic dependency structure
- Unified framework explaining the empirical pattern
- Traditional policies cannot resolve the tension; redistributing childcare to husbands can under certain conditions
- Quantitative analysis: both gender-neutral and gender-biased technological change matter
- Dynamic propagation through human capital amplifies effects across generations